

The Final Output

The potential manager would receive:

- A table recommending the number of bikes to place at each station at the start of the day (for a chosen fleet size).
- Plots of estimated pickup and return intensity functions (NHPP $\lambda(t)$) for each station.
- Simulation summaries, including expected shortages or overflow under a proposed placement.
- A results/folder containing example outputs (plots + tables) for at least three fleet sizes.

All outputs will be produced automatically from historical trip data ([sample_bike.csv](#) in our case).

High-Level Pipeline

1. Load historical trip data (start station, end station, timestamps).
2. Preprocess data (convert timestamps, group by station, filter days).
3. Estimate NHPP intensities for pickups and returns at each station.
4. Simulate daily demand using those estimated intensities (multiple runs).
5. Evaluate performance of a given initial bike allocation.
6. Search for a good allocation.
7. Generate final recommendations and save tables + plots.

The Functions Used

estimation.R

[estimate_arrival_rates\(data\)](#)

- Inputs:
 - data (data.frame): with columns start_station, end_station, start_time, end_time
- Outputs:
 - data.frame with columns start_station, end_station, hour, avg_trips, avg_avail, mu_hat
- Purpose:
 - Computes hourly *arrival rates* $\hat{\mu}$ for trips from each origin to each destination, using:
 - average number of hourly trips,
 - estimated time a station has bikes available,

- $\hat{\mu} = \text{avg_trips} / \text{avg_avail}$ when $\text{avg_avail} > 0$.
These $\hat{\mu}$ values form the basis for constructing NHPP intensity functions in the simulation step.

simulation.R

`simulate_station_day(lambda_pickup, lambda_return, seed = NULL)`

- Inputs:
 - two NHPP intensity functions
- Outputs:
 - simulated pickup and return times for one day
- Purpose:
 - Use $\hat{\mu}$ or $\lambda(t)$ to generate daily activity for one station.

`simulate_network(intensity_list, n_sim=100)`

- Inputs:
 - list of intensity functions per station
- Outputs:
 - list of simulated network-wide demand scenarios
- Purpose:
 - Perform repeated simulations needed for placement scoring.

placement.R

`score_placement(simulation, placement_vec)`

- Inputs:
 - simulated demand + vector of initial bikes
- Outputs:
 - numeric score (expected shortages/errors)
- Purpose:
 - Evaluate bike placements.

`search_best_placement(intensity_list, n_bikes)`

- Inputs:
 - intensity estimates + total fleet size
- Outputs:
 - suggested placement vector
- Purpose:

- Heuristic search for an effective distribution.

`generate_recommendations(intensity_list, fleet_sizes)`

- Inputs:
 - intensity list + vector of fleet sizes
- Outputs:
 - results saved to the **results** folder
- Purpose:
 - Generate tables + plots for the manager.

utils.R

`plot_intensity(intensity, station_id)`

Creates and saves NHPP intensity plots.

`plot_recommendation_table(tbl)`

Formats and exports placement tables.

Function Dependency Diagram

`estimate_arrival_rates()`



(build $\lambda(t)$ functions)



`simulate_network()`



`score_placement()`



`search_best_placement()`



`generate_recommendations()`

Refinements

Arrival rate estimation is grounded in actual trip + availability information, integrating naturally into NHPP modeling. NHPP intensities $\lambda(t)$ will be derived from $\mu\hat{t}$ for simulation. Simulation and placement scoring remain modular for unit testing. The structure allows easy addition of more complex optimization later.